

SYNTAX VERIFIER 2/24/67 (STV,32)

/SYNTAX VERIFIER VARIABLES

SUA=STEM+5
 SUD=STEM+7
 SJE=STEM+10
 SJF=STEM+11
 SUG=STEM+12
 SUK=STEM+13
 SUP=STEM+14
 SUQ=STEM+15
 SUR=STEM+16
 SUS=STEM+17
 SUT=STEM+20

/STV IOT AND SUBR

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STV,      LAR IRI
,STV,      LIO I (USERAC)
,STVJ,     DIO D
           DAP I (BTEM)
           LAC I (USERPC)
           DAC I (BTEM+1)
           LAC I (FSA)
           DAC G           /TEXT PTR
           DAC I (SUD)
           LAC B0
           DAC I (SUT)     /MULTIPLIER
           LAC I (SUQ)
           DAC I (SUK)
           DAC C           /OUTPUT PTR
           LAC I (SUP)
           DAC I (SUF)     /PUSHDOWN PTR

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/MAIN LOOP FOR INTERPRETATION

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STVHL,     LIO I D           /GET PSEUDO INSTR
           ISP I (SUT)
           IDX D           /IDX PC CONTINGENT ON MULTIPLIER
           DIO E           /SAVE CURRENT PSEUDO INSTR
           CLA
           RCL 3S
           ADD (LAC STVDTR)
           DAC A
B2,         XGT A           /CONSTANT 100000, DISPATCH LOOKUP
           SEL I
           IDA           /ADD 1 IF IN WIN MODE
           DAC A
           CLA+USWP+UCLF 7   /CLEAR ALL FLAGS EXCEPT LINK
           RCL 3S
           IFI           /SET FLAGS 4,5,6 FROM BITS 3,4,5
           LIO E
           JMP A           /DISPATCH JMP

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/IDX OPCODE

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STV44,     DZM I (BTEM+5) /GROUP COUNTER
           JSP STVTRA      /GET TEXT PTR INTO E
           LAC G
           DAC I (BTEM+4) /SPACE FOR RESTORING
           DAC I (BTEM+2) /FARTHEST SUCCESSFUL SO FAR
STV4A,     JSP STVLE      /LCH I E; DAC A; SKIP ON NOT EQM
           JMP STV4C      /EQM; SUCCEEDED
STV4A1,    LAC G
           SAD I (BTEM+2)
           STF 1          /THIS WILL BE A NEW RECORD IF IT WINS
           JSP ATM        /LCH I G; DAC F
           LAC A
           SAD F
           JMP STV4A      /CHAR'S AGREE, LOOP
STV4R,     JSP STVLE      /FAILURE, WAIT FOR EQM
           JMP STV4D      /FOUND, TRY NEXT CASE
           JMP STV4B

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STV4C,  SZF I 1          /SUCCESS; IS THIS A NEW RECORDS?
        JMP STV4D        /NO, IGNORE
        STF 2            /SET WIN FLAG
        LAC G
        DAC I (BTEM+2)   /REMEMBER HIS TEXT PTR
        LAC I (BTEM+5)
        DAC I CBTM3      /REMEMBER HIS GROUP
STV4D,  LAC I (BTEM+4)   /ROUTINE TO TRY NEXT CASE
        DAC G
        CLF 1
        JSP STVLE
        JMP STV4D1       /2 EOM'S IN A ROW; NEW GROUP
        JMP STV4A1       /GO INTO CHAR. LOOP
STV4D1, IDX I (BTEM+5)
        JSP STVLE
        JMP STV4E        /3 EOM'S IN A ROW; FINISHED
        JMP STV4A1

STV4E,  SZF I 2          /WAS THERE A WINNER?
        JMP STVLOS
        LAC I (BTEM+2)   /YES, REMEMBER HIS PTR
        DAC G
        LAC I CBTM3      /SAVE HIS GROUP #
STV4A,  DAC I C
        IDX C
STV4,   JMP STVML        /ALL DONE; ALSO LOSE MODE
        SZF 4
        JMP STV44        /IDX OPCODE
        LAI              /?
        SCH+UIDA         /ADD OPCODE, CALCULATE MULTIPLIER
        DAC I (SUT)
STV4,   JMP STVML
        JSP ATM
        JSP REATM        /CHECK CHAR, AGAINST ATOM TABLE
        JMP STVML        /OK
STVLOS, CLL+UCML        /NG, ENTER LOSE MODE
        LAC P0
        DAC I (SUT)      /Clobber MULTIPLIER
STV5,   JMP STVML
STV5L,  JSP ATM
        JSP REATM        /CHECK CHAR, AGAINST ATOM TABLE
        JMP STV5L        /OK, KEEP CYCLING
STV5UL, LAC I (SUD)
        IDC
        DAC E
STV5UM, LAW I 1
        ADD G
        IDC
        IDC
        DAC G
        IDC
        SAD E
        JMP STVML
        LCH G
        SAD ML6
        JMP STV5UM

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STV1,    JMP STVML          /GET TEXT PTR
          JSP STVTRA
          LAC G
          DAC I (FSA)       /UPDATE FSA SO IT CAN BE REFERRED TO
          LAC I E           /GET ARG
          SZF 4
          JMP STV14         /DCH OPCODE; SAVE ARG
STV1L,    JSP STVLE
          JMP STVML         /EOM, XCT OPCODE SUCCEEDED
          JSP ATM
          LAC A
          SAD F
          JMP STV1L         /KEEP INSISTING ON EXACT MATCH
          JMP STVLOS        /NO MATCH, FAIL

STV2,    JMP STV2L          /LAC IN LOSE MODE
          LAW 2             /LAC IN WIN MODE
          AND E
          SZA
          JMP STV2SP        /SUPER POP
STVPOP,   LAC I (SUF)       /NORMAL POP
          SAD I (SUP)
          JMP STVNPP        /TOP LEVEL, CAN'T POP
          SUB (3)
          DAC I (SUF)       /UPDATE PUSHDOWN PTR
          DAC A             /TEMPORARY PUSHDOWN PTR
          LAC I A
          DAC I (SUD)       /POP SUD
          IDX A
          LAC I A
          CLI               /PREPARE TO UNPACK POP SUT;SUK; PSEUDO PC
          SCR 6S
          SZL
          CLA               /NO MULTIPLIER IN LOSE MODE
          XOR B0
          DAC I (SUT)
          RIL 6S
          IDX A
          LAC I A
          RCL 6S
          DIO I (SUK)
          RAP 6S            /PSEUDO PC TO AC
          JMP STV7I         /PSEUDO GO

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STVNFP,   LAC I (BTEM)
          SZL
          JMP STVRUR      /GIVE R1 ON CAN'T LOSE POP
          JSF ATM
          LAC F
          SAS ML4
          JMP STVWL      /NOT EOM, LOSE THEN LAC
STV2SP,   IOX I (BTEM)   /GIVE R2
STVRUR,   DAP H          /SUBR TO RESET USER'S REGISTERS
          LAC C
          DAC I (SUG)    /OUTPUT PTR
          LAC D
          DAC I (SHA)    /PSEUDO PC
          LAC G
          DAC I (FSA)    /TEXT PTR
          LAC I (BTEM+1)
          DAC I (USERPC) /REAL PC (SOMETIMES REDUNDANT)
          JMP HEXIT

/DON'T LAW IN LOSE MODE
STV7L,    SZF I 4
          JMP STV7II
          DEM E          /LOOK LIKE LAC 0
/WIN ENTRY TO LAC IN LOSE MODE
STVWL,    CLL UCML      /ENTER LOSE MODE
STV2L,    LAC I (SUK)    /RESTORE PTRS TO BEG OF LEVEL
          DAC C
          LAC I (SUD)
          DAC G
          LAC E
          SAD CLACH      /((LAC H)
          JMP STV2SP
          LAW 5
          AND E
          SAS B17        /STAY IN LOSE MODE ON LAC 1
          CLL            /ENTER WIN MODE ON LAC 0; LAC 5
          LIO B17
          DIO E          /LOOK LIKE LAC 1 ON RECYCLE (LAC 5)
          SZA
          JMP STVPOP      /POP ON LAC 1; LAC 5

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STV3,      JMP STVML
            LAC I (SUF)      /PUSH OPCODE (DIP)
            SAD I (SUE)
            JMP STVLOS      /CAN'T PUSH, LOSE
            DAC A           /TEMPORARY PUSHDOWN PTR
            LAC I (SUD)      /PUSH SUD
            DAC I A
            IDX A
            LAC I (SUT)      /PACK PUSH SUT; SUK; PSEUDO PC
            LIO I (SUK)
            RIL 6S
            RCL 6S
            DAC I A
            IDX A
            LAC D
            RCL 6S
            RAR 6S
            DAC I A
            IDX A
            DAC I (SUF)      /UPDATE PUSHDOWN PTR
            LAC C
            DAC I (SUK)      /UPDATE "BEGINNING OF LEVEL" PTRS
            LAC G
            DAC I (SUD)
            LAC B0
            DAC I (SUT)      /CLOBBER MULTIPLIER
            JMP STV7II       /NOW DO A "LAW" OPCODE

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STV7,      JMP STV7L
STV7II,    JSP STVTRA
            LAC E
STV7I,     DEX D
            DAF D
STV6,      JMP STVML
            JSP STVTRA
            JSP STVRUR
            LAC E
            JMP GO

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/STVL AND STVW IOTS

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STVL,      CHL
            LAC B0
            DAC I (SUT)
            LAC I (FSA)
            DAC G
            LAC I (SUG)
            DAC C
            LAC I (SUA)
            JMP STV7I

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/UNRESTORE USER REGISTERS

/ATOM TABLE SUBROUTINES

/ENTRY TO GET A CHAR, AND SKIP ALL IRRELEVANT (SUS) CHAR,S

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ATM,      SUB B17
          DAP H
          LIO I (SUS)      /PUT MASK IN IO
          LQH I G
          SAD ML4
          JMP ATMW          /WARNING
ATMR0,    DAC F            /SAVE CHAR,
          RAL 6S
ATMI,     ADD (LAC ATOM)
          DAC B
          XOT B            /LOOK UP ATOM TABLE ENTRY
          SZA I
          JMP XHEXIT        /R2, EOM ALWAYS FAILS
ATMNC,    DIO B            /SAVE MASK
          RIL 3S           /PUT CLASS BIT IN SIGN OF IO
          XAI
          SPA
          JMP ATMD          /DIFFERENT CLASSES
          XAI              /RESTORE AC
          AND B
          AND (37777)
          SZA I
XHEXIT,   IDX H            /NO BITS THE SAME, FAIL (R2)
          JMP HEXIT        /OK, R1
ATMD,     RIL 1S           /MOVE "OTHER CLASS" BIT INTO SIGN OF IO
          SPI I
          IDX H
          JMP HEXIT
ATMW,     LQH I G
          SAD ML4
          JMP ATMR0        /TREAT RUBOUT LIKE EOM
          RAR 6S
          IOR ML6
          DAC F            /12BIT CHAR, IN F
ATMWI,    LAC (404000)     /ATOM TABLE ENTRY FOR 12 BIT CHAR,S
          JMP ATMNC

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/ENTRY TO TEST CHAR, IN F AGAINST MASK IN E

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REATM,    DAP H
          CLA
          LIO F
          ROL 6S
          LIO E
          SNI
          SAS (74)
          JMP REATM1
          JMP STV5UL
REATM1,   SAS (77)
          JMP ATMI
          JMP ATMWI

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/ROUTINE TO LCH I E, DAC A AND SKIP ON NOT EOM

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STVLE,   DAP H
         LCH I E
         SAD ML6
         JMP STVLEW      /WARNING
STVLEI,  DAC A
         SAS ML4
         IDX H           /NOT EIM
         JMP HEXIT
STVLEW,  RCL 6S
         LCH I E
         RCR 6S
         JMP STVLEI      /12 BIT CHAR. IN A

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/RELOCATABLE TRACING SUBROUTINE FOR STV

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STVTRA,  DAP H
         LAW 7777        /TOP 6 BITS 0 FIRST TIME THROUGH
         AND E
         LIO E
STVTRL,  RIL 4S
         SPI
         ADD I (SUR)
         DAC E
         RIL 1S
         SPI I
         JMP HEXIT
         LAC I E
         AND (607777)
         LIO I E
         JMP STVTRL

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/OPCODE DISPATCH TABLE

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STVDTB,  JMP STV0
         JMP STV1
         JMP STV2
         JMP STV3
         JMP STV4
         JMP STV5
         JMP STV6
         JMP STV7

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/ATOM TABLE, SIGN BIT HAS CLASS = 0 MEANS EOM

ATOM, B13, 20		/SPACE
	400010	/!
B5,	10000	/"
	410000	/#
	410000	/\$
	410000	/%
	410000	/8
B8,	1000	/'
	400020	/(
	400040	/)
	400001	/*
B11,	100	/+
B7,	2000	/,
B12,	40	/=
CJORG,		
B10,	200	/.
	400002	//
B16,	REPEAT 10,, 2	/DIGITS
	400004	/i
B6,	4000	/i
	400400	/<
	401000	/F
	402000	/>
B14,	10	/?
ATOM+40,	410000	/@
B9,	400	/A
B17,	REPEAT 3, 1	/BCD
400	REPEAT 3, 1	/E FGH
400	REPEAT 5, 1	/I JKLMN
400	REPEAT 5, 1	/O PQRST
400	REPEAT 3, 1	/U VWX
400	1	/Y Z
	400100	/E
	0	/EOM
	400200	/J
B15,	4	/CRLF
ATOM+77,	/NO ENTRY FOR 77	

START